

Learning Phase Competition for Traffic Signal Control

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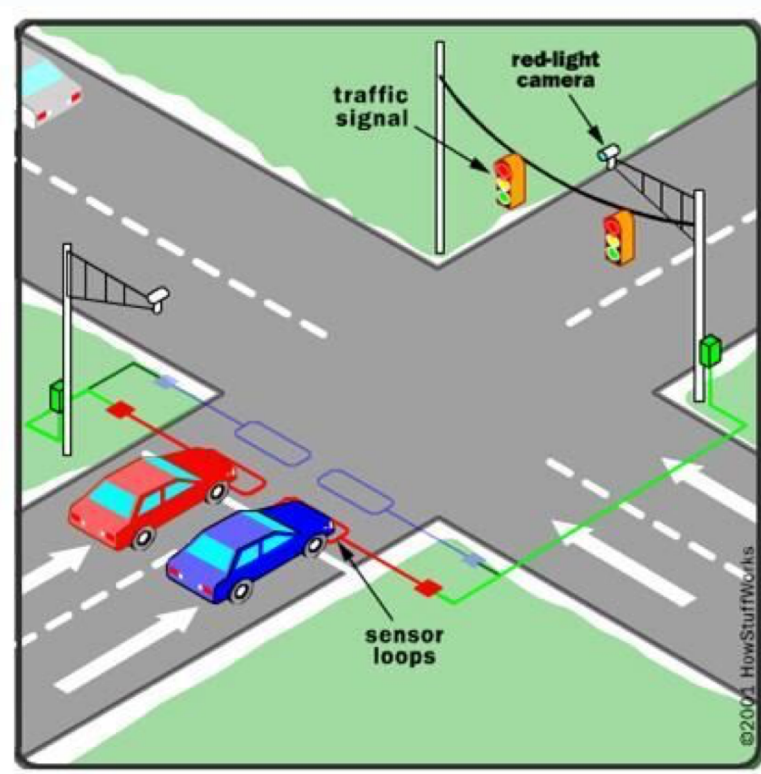
Motivation

Traffic congestion causes additional travel time

	New York	Paris	Tokyo	London
MORNING PEAK	55%	71%	64%	62%
EVENING PEAK	69%	67%	63%	65%

Data and images from
https://www.tomtom.com/en_gb/traffic-index/

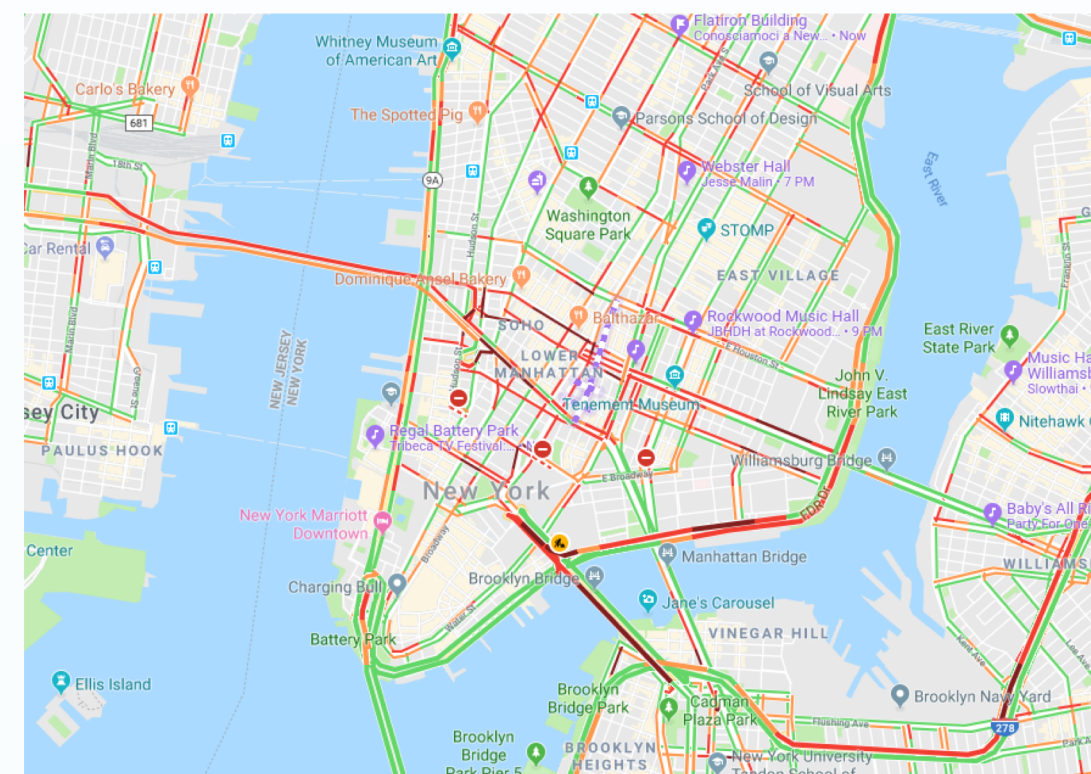
Problem: how to use traffic data to control the signal?



Loop sensor



Camera

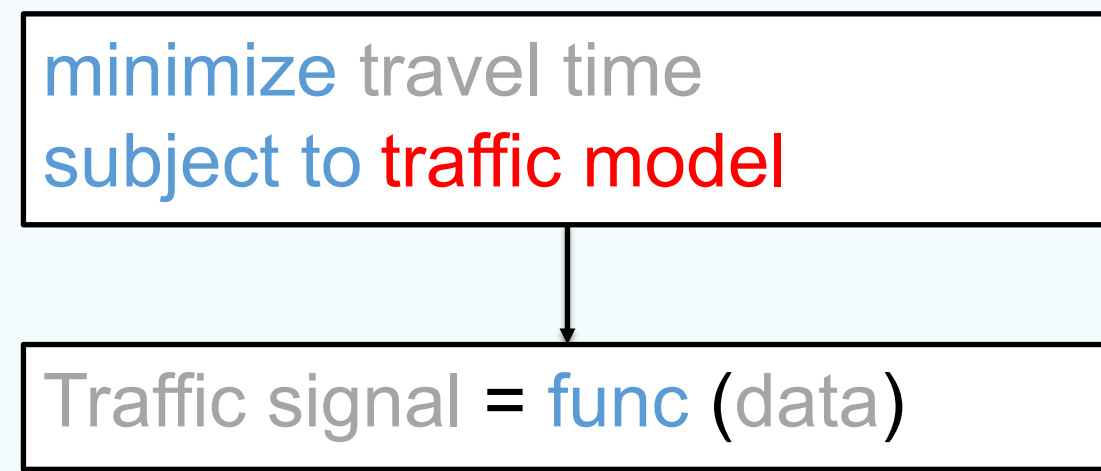


Speed

Image from Google Maps

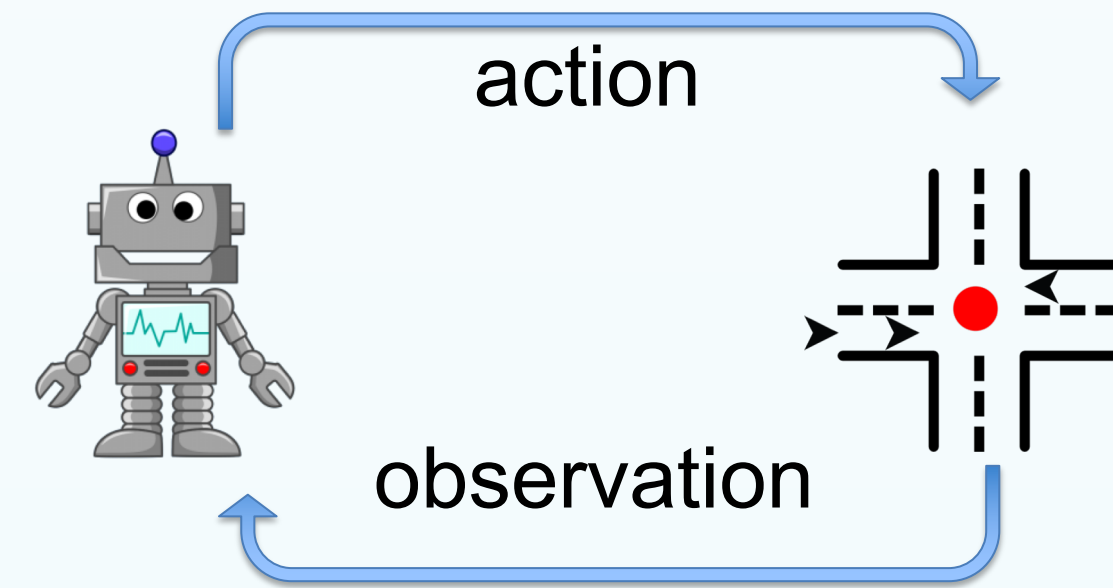
Related work & Preliminary

Transportation Methods
Decision based on assumptions



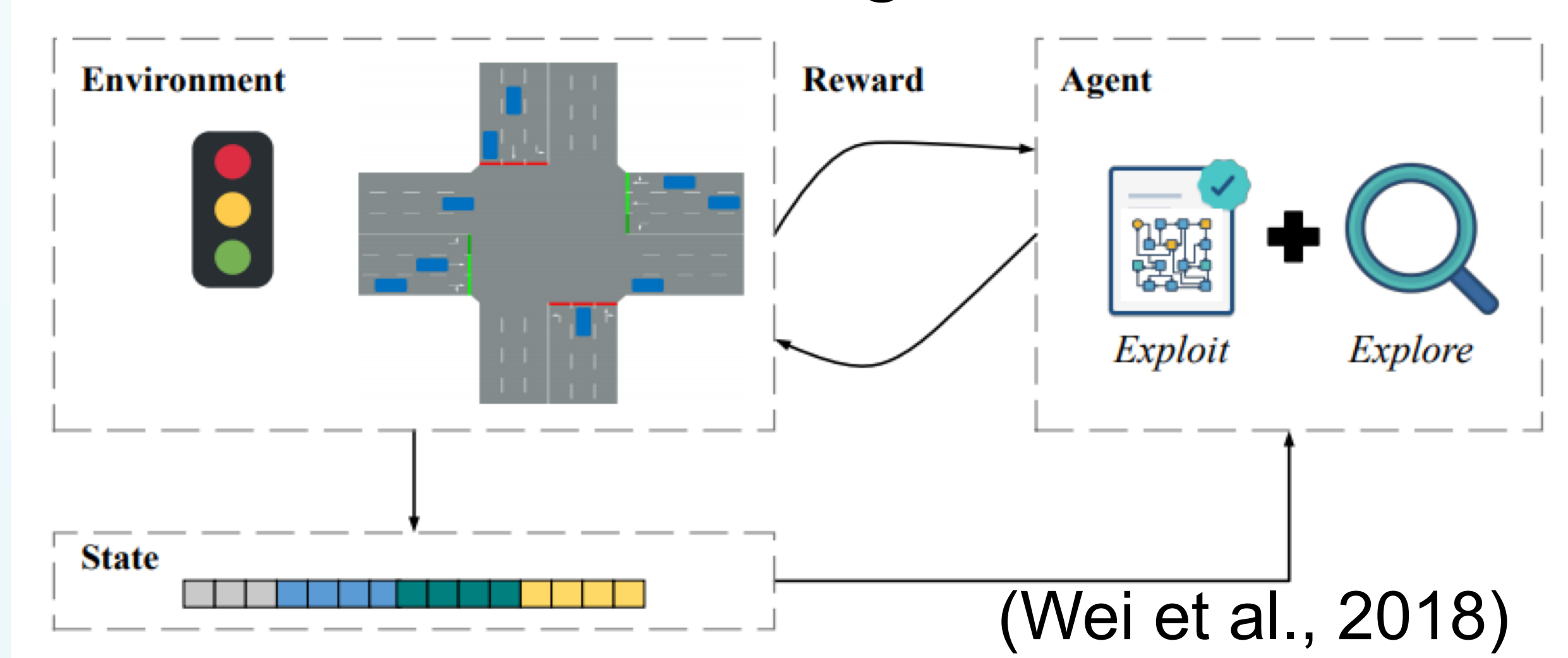
Decision = func(data) is static

Data Driven Methods
Directly learning from real-world data



Decision = func(data) is online learned

A framework of RL for Traffic Signal Control



(Wei et al., 2018)

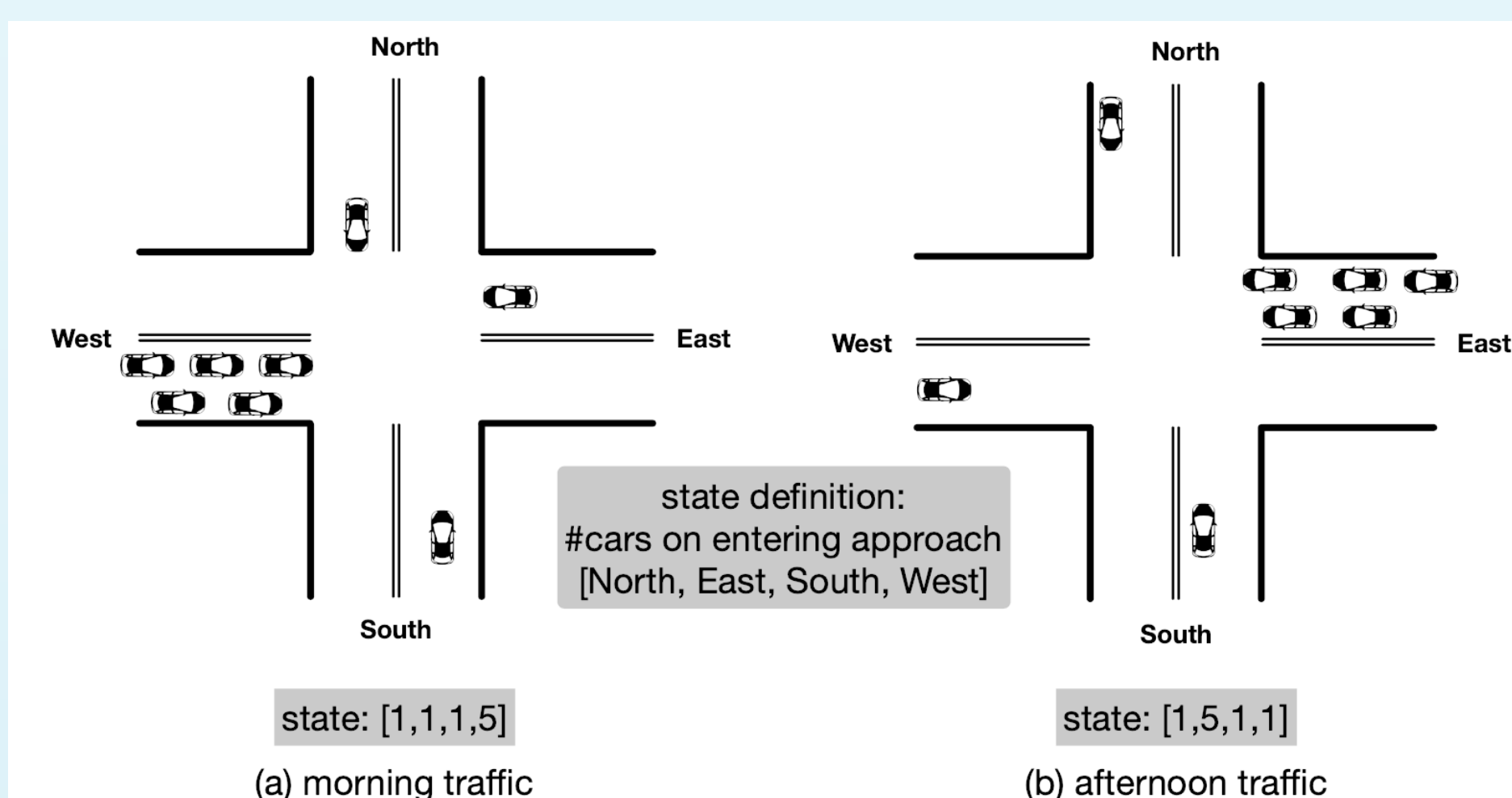
- Reward: queue length
- Action: signal phase for next time interval
- State: number of vehicles

Previous works in RL for Traffic Signal Control

Problem design Method

- State
- Reward
- Action
- Value-based v.s. policy based
- Q-learning, policy gradient, actor-critic
- Model-based v.s. model-free

Issues with previous works



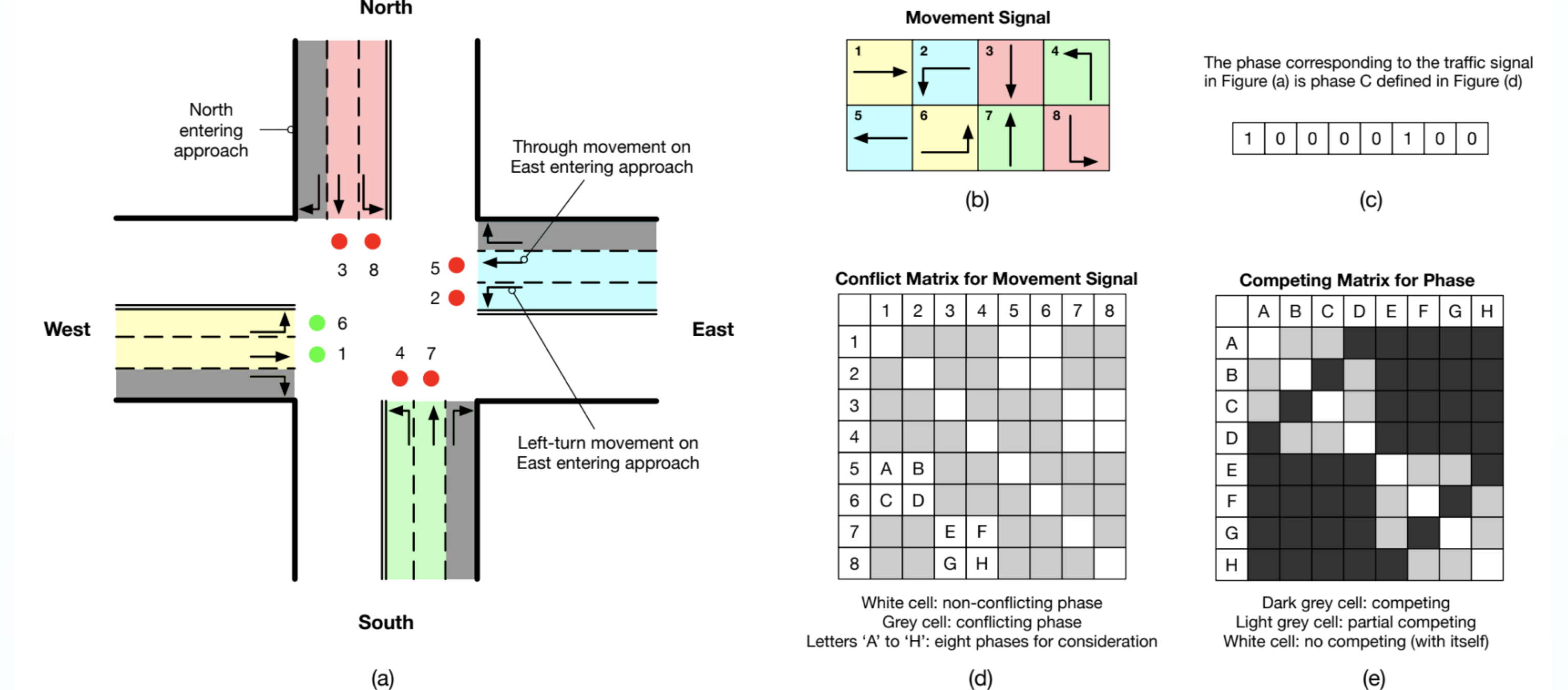
Similar cases are considered repetitively.

Algorithms will struggle in multiple-phase scenarios with larger state space!

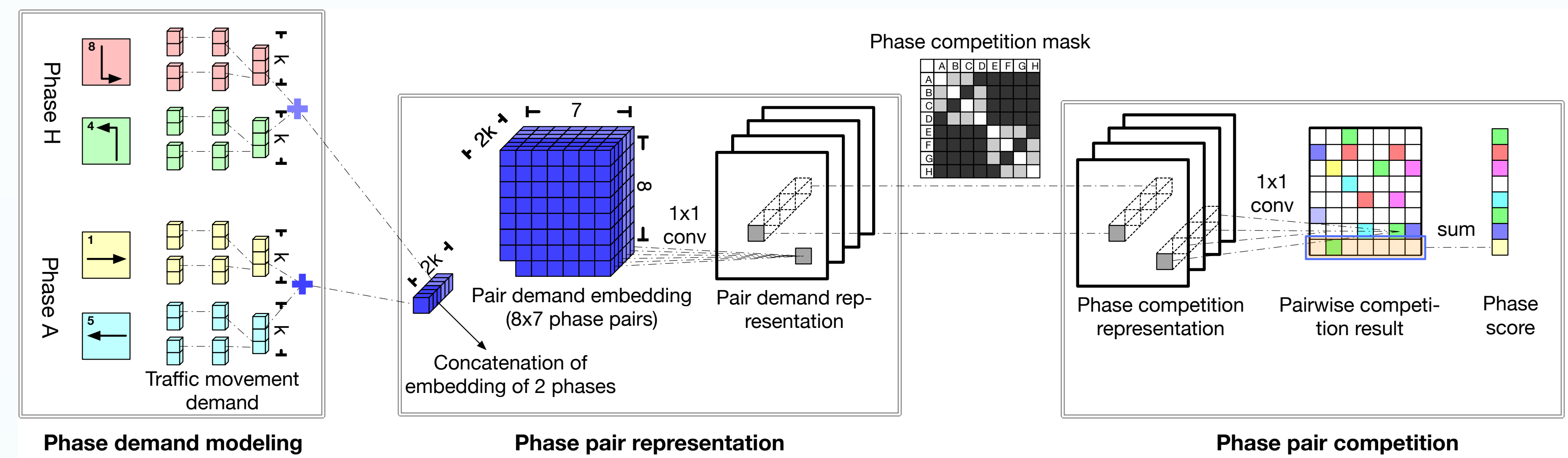
Methodology

Key idea

- (1) Combine the compatible traffic movements into signal phase.
- (2) Model the competition between pair of phases.



Model framework



Experimental Results

Table 1: Overall performance. Travel time is reported in the unit of second.

Model	Jinan							Hangzhou					
	1	2	3	4	5	6	7	1	2	3	4	5	6
Fixedtime	118.82	250.00	233.83	297.23	101.06	104.00	146.66	271.16	192.32	258.93	207.73	259.88	237.77
Formula	107.92	195.89	245.94	159.11	76.16	100.56	130.72	218.68	203.17	227.85	155.09	218.66	230.49
SOTL	97.80	149.29	172.99	64.67	76.53	92.14	109.35	179.90	134.92	172.33	119.70	188.40	171.77
DRL	98.90	235.78	182.31	73.79	66.40	76.88	119.22	146.50	118.90	218.41	80.13	120.88	147.80
IntelliLight	88.74	195.71	100.39	73.24	61.26	76.96	112.36	97.87	129.02	186.04	81.48	177.30	130.40
A2C	135.81	166.97	226.82	43.28	67.05	148.69	236.17	110.91	98.56	187.41	86.56	116.70	128.88
FRAP	66.40	88.40	84.32	33.83	54.43	61.72	72.31	80.24	79.43	110.33	67.87	92.90	88.28
Improvement	25.17%	40.79%	16.01%	47.69%	11.15%	19.72%	33.87%	18.01%	33.20%	35.98%	15.30%	23.15%	32.30%

Table 3: Performance on different intersection structures.

Model	3-approach	4-approach	5-approach
Fixedtime	166.16	93.21	211.26
Formula	159.12	67.17	231.33
SOTL	123.43	65.39	124.71
DRL	108.94	125.84	140.33
IntelliLight	108.27	60.38	151.92
A2C	132.56	52.64	172.60
FRAP	81.57	48.83	110.66

Table 4: Performance in a multi-intersection environment.

Model	Jinan	Hangzhou	Atlanta
Fixedtime	880.18	823.13	493.49
Formula	385.46	629.77	831.34
SOTL	1422.35	1315.98	721.15
DRL	1047.52	1683.05	769.46
IntelliLight	358.83	634.73	306.07
A2C	316.61	591.14	244.10
FRAP	293.35	528.44	124.42



Images from Google Maps

Acknowledgements

The work was supported in part by NSF awards #1652525, #1618448 and #1639150. The views and conclusions contained in this paper are those of the authors and should not be interpreted as representing any funding agencies.

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We are implementing our model in Hangzhou China!

Try to find more related researches? Just scan QR code on the right or visit <https://traffic-signal-control.github.io>

